

SmartLogger Modbus SCADA Integration Guide

Device Target: SmartLogger V300R024C10SPC161 Base Layer Profiles

Protocol Baseline: Modbus-TCP / Ethernet Standard

1. Communication Protocol Layer Profiles

For implementation of custom driver instances or SCADA platform adapters, programmers must align structural formatting parameters to the following runtime profiles:

Network Property	Implementation Specification Target
Transmission Medium	Ethernet Base Connection Layer
Default Transport Port	502
Data Encoding Structure	Big-Endian Encoding Format (Most Significant Byte First)
Logic Device Target ID	Fixed to 0 (SmartLogger Local / Aggregated Fleet Context)
Transaction Execution Bound	5.0 Seconds Framework Boundary Limits for Complete Timed-Out Triggers

2. Supported Modbus Function Codes

The system driver engine exposes data structures via a specific subset of standard Modbus Application Layer codes. Interrogation routines must restrict commands to these functions:

Function Code	Operation Definition	Application Range Details
0x03	Read Holding Registers	Supports continuous batch execution patterns for single or multiple data blocks.
0x06	Write Single Register	Supports selective targeted modification writes for isolated variables.
0x10	Write Multiple Registers	Supports atomic block updates and structured configuration setups for array ranges.

3. High-Priority SCADA Telemetry Register Matrix

The following parameters are essential for tracking operational health, power limits, total production yield, and current alarm states:

Parameter Target Name	Address	Quantity	Type	Gain	Unit	Application & Driver Constraint Logic
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Active adjustment (Volatile)	40420	2	U32	10	kW	Adjusts total active power. This register is volatile; settings are cleared upon a device power cycle.
Active adjustment (Fail-safe)	40424	2	U32	10	kW	Adjusts total active power. This register is persistent and fail-safe; configurations remain valid through cycles.
Reactive adjustment	40422	2	I32	10	kVar	Adjusts absolute total plant reactive power. Minimum 1-second interval execution delays required between consecutive adjustments.
Active power	40525	2	I32	1000	kW	Real-time aggregated active power output measured across all operating inverters in the grid network.
Reactive power	40544	2	I32	1000	kVar	Real-time aggregated reactive power output measured across all active on-grid elements.
Input power	40521	2	U32	1000	kW	Total real-time input power aggregated from connected DC array strings.
Power factor	40532	1	I16	1000	N/A	Total concurrent grid connection power factor reading.
E-Daily	40562	2	U32	10	kWh	Daily energy accumulation yield tracker. Reset bounds follow localized controller midnight parameters.
E-Total	40560	2	U32	10	kWh	Lifetime accumulated energy production profile yield generated since system commissioning.
Plant status	40543	1	U16	1	N/A	System tracking profiles. Mappings: 1:Unlimited, 2:Limited, 3:Idle, 4:Outage (Fault), 5:Communication interrupt.
Alarm Info 1	50000	1	U16	1	N/A	Aggregated fault telemetry mask (Bit 3: Abnormal Active Schedule, Bit 11: Abnormal Reactive Schedule).
Alarm Info 2	50001	1	U16	1	N/A	Aggregated fault telemetry mask (Bit 1: MCB Disconnect, Bit 2: Abnormal Cubicle, Bit 3: Address Conflict).

4. Driver Integration & Programming Rules

- **Atomic Handling:** For 2-quantity multi-word parameters (such as U32 or I32 types spanning two sequential registers), the client driver application must perform consecutive block queries within a single transaction to prevent half-word context mismatches.
- **Scaling Applications:** Collected integers must be mathematically multiplied by the respective scale reciprocal (multiplied by 1/Gain) to normalize accurate engineering units inside SCADA databases.
- **Adjustment Guardrails:** Writes directed to both active and reactive power adjustment blocks require strict enforcement of a minimum 1.0-second cooldown period between successive transfers. Input values exceeding absolute boundaries (queried from maximum active boundary parameter 40697) are instantly rejected by the SmartLogger controller logic.